



GyaanVault

SPPU M.Sc. Data Science — Free Study Material

Statistics for Data Science

DS-501-MJ

Sample / Practice Question Paper

Semester I · Core · Practice Paper

Time: 3 Hours

Maximum Marks: 70

Instructions: (1) All questions carry equal weightage where indicated. (2) Figures to the right indicate full marks. (3) Draw neat diagrams / write code wherever necessary.

Q1. Answer any FIVE of the following. [5 × 2 = 10]

- a) Define mean, median and mode.
- b) What is standard deviation?
- c) State the range of the correlation coefficient.
- d) Define skewness.
- e) State Bayes' theorem.
- f) What is a p-value?
- g) Differentiate correlation and regression.

Q2. Answer the following. [15]

A) Explain measures of central tendency and dispersion with examples.

OR

B) Explain Karl Pearson's coefficient of correlation and its interpretation.

Q3. Answer the following. [15]

A) Explain simple and multiple linear regression. How is a model evaluated using R^2 and adjusted R^2 ?

OR

B) Explain logistic regression and how it differs from linear regression.

Q4. Answer the following. [15]

A) Describe hypothesis testing. Explain t-test, chi-square test and ANOVA.

OR

B) Explain sampling techniques: simple random and stratified random sampling.

